



Advanced Linear Algebra: Foundations to Frontiers

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2.5.2 Summary ¶

Given $x, y \in \mathbb{C}^m$

- their dot product (inner product) is defined as

$$x^H y = \overline{x}^T y = \overline{x^T y} = \overline{\chi_0} \psi_0 + \overline{\chi_1} \psi_1 + \cdots + \overline{\chi_{m-1}} \psi_{m-1} = \sum_{i=0}^{m-1} \overline{\chi_i} \psi_i.$$

- These vectors are said to be orthogonal (perpendicular) iff $x^H y = 0$.
- The component of y in the direction of x is given by

$$\frac{x^H y}{x^H x} x = \frac{x x^H}{x^H x} y.$$

The matrix that projects a vector onto the space spanned by x is given by

$$\frac{x x^H}{x^H x}.$$

- The component of y orthogonal to x is given by

$$y - \frac{x^H y}{x^H x} x = \left(I - \frac{x x^H}{x^H x} \right) y.$$

Thus, the matrix that projects a vector onto the space orthogonal to x is given by

$$I - \frac{x x^H}{x^H x}.$$

Given $u, v \in \mathbb{C}^m$ with u of unit length

- The component of v in the direction of u is given by

$$u^H v u = u u^H v.$$

- The matrix that projects a vector onto the space spanned by u is given by

$$u u^H$$

- The component of v orthogonal to u is given by

$$v - u u^H v = (I - u u^H) v.$$

- The matrix that projects a vector onto the space that is orthogonal to x is given by

$$I - uu^H$$

- Let $u_0, u_1, \dots, u_{n-1} \in \mathbb{C}^m$. These vectors are said to be mutually orthonormal if for all $0 \leq i, j < n$

$$u_i^H u_j = \begin{cases} 1 & \text{if } i = j \\ 0 & \text{otherwise} \end{cases}.$$

- Let $Q \in \mathbb{C}^{m \times n}$ (with $n \leq m$). Then Q is said to be

- an orthonormal matrix iff $Q^H Q = I$.
- a unitary matrix iff $Q^H Q = I$ and $m = n$.
- an orthogonal matrix iff it is a unitary matrix and is real-valued.

- Let $Q \in \mathbb{C}^{m \times n}$ (with $n \leq m$). Then $Q = \left(q_0 \mid \dots \mid q_{n-1} \right)$ is orthonormal iff $\{q_0, \dots, q_{n-1}\}$ are mutually orthonormal.

Definition 2.5.2.1. Unitary matrix. Let $U \in \mathbb{C}^{m \times m}$. Then U is said to be a unitary matrix if and only if $U^H U = I$ (the identity).

- If $U, V \in \mathbb{C}^{m \times m}$ are unitary, then

- $U^H U = I$.
- $U U^H = I$.
- $U^{-1} = U^H$.
- U^H is unitary.
- UV is unitary.

- If $U \in \mathbb{C}^{m \times m}$ and $V \in \mathbb{C}^{n \times n}$ are unitary, $x \in \mathbb{C}^m$, and $A \in \mathbb{C}^{m \times n}$, then

- $\|Ux\|_2 = \|x\|_2$.
- $\|U^H A\|_2 = \|UA\|_2 = \|AV\|_2 = \|AV^H\|_2 = \|U^H AV\|_2 = \|UAV^H\|_2 = \|A\|_2$.
- $\|U^H A\|_F = \|UA\|_F = \|AV\|_F = \|AV^H\|_F = \|U^H AV\|_F = \|UAV^H\|_F = \|A\|_F$.
- $\|U\|_2 = 1$
- $\kappa_2(U) = 1$

- Examples of unitary matrices:

- Rotation in 2D: $\begin{pmatrix} c & -s \\ s & c \end{pmatrix}$.
- Reflection: $I - 2uu^H$ where $u \in \mathbb{C}^m$ and $\|u\|_2 = 1$.

Change of orthonormal basis: If $x \in \mathbb{C}^m$ and $U = \left(u_0 \mid \cdots \mid u_{m-1} \right)$ is unitary, then

$$x = (u_0^H x)u_0 + \cdots + (u_{m-1}^H x)u_{m-1} = \left(u_0 \mid \cdots \mid u_{m-1} \right) \underbrace{\begin{pmatrix} u_0^H x \\ \vdots \\ u_{m-1}^H x \end{pmatrix}}_{U^H x} = UU^H x.$$

Let $A \in \mathbb{C}^{n \times n}$ be nonsingular and $x \in \mathbb{C}^n$ a nonzero vector. Consider

$$y = Ax \quad \text{and} \quad y + \delta y = A(x + \delta x).$$

Then

$$\frac{\|\delta y\|}{\|y\|} \leq \underbrace{\|A\| \|A^{-1}\|}_{\kappa(A)} \frac{\|\delta x\|}{\|x\|},$$

where $\|\cdot\|$ is an induced matrix norm.

Theorem 2.5.2.2. Singular Value Decomposition Theorem. Given $A \in \mathbb{C}^{m \times n}$ there exist unitary $U \in \mathbb{C}^{m \times m}$, unitary $V \in \mathbb{C}^{n \times n}$, and $\Sigma \in \mathbb{R}^{m \times n}$ such that $A = U\Sigma V^H$. Here $\Sigma = \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right)$ with

$$\Sigma_{TL} = \begin{pmatrix} \sigma_0 & 0 & \cdots & 0 \\ 0 & \sigma_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{r-1} \end{pmatrix} \quad \text{and} \quad \sigma_0 \geq \sigma_1 \geq \cdots \geq \sigma_{r-1} > 0.$$

The values $\sigma_0, \dots, \sigma_{r-1}$ are called the singular values of matrix A . The columns of U and V are called the left and right singular vectors, respectively.

Let $A \in \mathbb{C}^{m \times n}$ and $A = U\Sigma V^H$ its SVD with

$$U = \left(U_L \mid U_R \right) = \left(u_0 \mid \cdots \mid u_{m-1} \right),$$

$$V = \left(V_L \mid V_R \right) = \left(v_0 \mid \cdots \mid v_{n-1} \right),$$

and

$$\Sigma = \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & 0 \end{array} \right), \quad \text{where } \Sigma_{TL} = \begin{pmatrix} \sigma_0 & 0 & \cdots & 0 \\ 0 & \sigma_1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \sigma_{r-1} \end{pmatrix} \quad \text{and} \quad \sigma_0 \geq \sigma_1 \geq \cdots \geq \sigma_{r-1} > 0.$$

Here $U_L \in \mathbb{C}^{m \times r}$, $V_L \in \mathbb{C}^{n \times r}$ and $\Sigma_{TL} \in \mathbb{R}^{r \times r}$. Then

- $\|A\|_2 = \sigma_0$. (The 2-norm of a matrix equals the largest singular value.)
- $\text{rank}(A) = r$.
- $\mathcal{C}(A) = \mathcal{C}(U_L)$.
- $\mathcal{N}(A) = \mathcal{C}(V_R)$.
- $\mathcal{R}(A) = \mathcal{C}(V_L)$.
- Left null-space of $A = \mathcal{C}(U_R)$.
- $A^H = V \Sigma^T U^H$.
- SVD: $A^H = V \Sigma U^H$.
- Reduced SVD: $A = U_L \Sigma_{TL} V_L^H$.
- $$A = \underbrace{\sigma_0 u_0 v_0^H}_{\sigma_0} + \underbrace{\sigma_1 u_1 v_1^H}_{\sigma_1} + \cdots + \underbrace{\sigma_{r-1} u_{r-1} v_{r-1}^H}_{\sigma_{r-1}}.$$
- Reduced SVD: $A^H = V_L \Sigma U_L^H$.
- If $m \times m$ matrix A is nonsingular: $A^{-1} = V \Sigma^{-1} U^H$.
- If $A \in \mathbb{C}^{m \times m}$ then A is nonsingular if and only if $\sigma_{m-1} \neq 0$.
- If $A \in \mathbb{C}^{m \times m}$ is nonsingular then $\kappa_2(A) = \sigma_0 / \sigma_{m-1}$.
- (Left) pseudo inverse: if A has linearly independent columns, then $A^\dagger = (A^H A)^{-1} A^H = V \Sigma_{TL}^{-1} U_L^H$.
- v_0 is the direction of maximal magnification.
- v_{n-1} is the direction of minimal magnification.
- If $n \leq m$, then $Av_j = \sigma_j u_j$, for $0 \leq j < n$.

Theorem 2.5.2.3. Given $A \in \mathbb{C}^{m \times n}$, let $A = U \Sigma V^H$ be its SVD. Assume the entries on the main diagonal of Σ are $\sigma_0, \dots, \sigma_{\min(m,n)-1}$ with $\sigma_0 \geq \dots \geq \sigma_{\min(m,n)-1} \geq 0$. Given k such that $0 \leq k \leq \min(m, n)$, partition

$$U = \left(U_L \mid U_R \right), V = \left(V_L \mid V_R \right), \text{ and } \Sigma = \left(\begin{array}{c|c} \Sigma_{TL} & 0 \\ \hline 0 & \Sigma_{BR} \end{array} \right),$$

where $U_L \in \mathbb{C}^{m \times k}$, $V_L \in \mathbb{C}^{n \times k}$, and $\Sigma_{TL} \in \mathbb{R}^{k \times k}$. Then

$$B = U_L \Sigma_{TL} V_L^H$$

is the matrix in $\mathbb{C}^{m \times n}$ closest to A in the following sense:

$$\|A - B\|_2 = \min_{\substack{C \in \mathbb{C}^{m \times n} \\ \text{rank}(C) \leq k}} \|A - C\|_2.$$

In other words, B is the matrix with rank at most k that is closest to A as measured by the 2-norm. Also, for this B ,

$$\|A - B\|_2 = \begin{cases} \sigma_k & \text{if } k < \min(m, n) \\ 0 & \text{otherwise.} \end{cases}$$